SPANS-06: Security Analysis with Spans

> **Series:** SPANS — Distributed Tracing and Spans | **Notebook:** 6 of 8 | **Created:** December 2025 | **Last Updated:** 10/28/2026

Protecting Distributed Traces and Ensuring Compliance

This notebook demonstrates how to use span data for security analysis, audit for sensitive data exposure, and ensure compliance with regulations.

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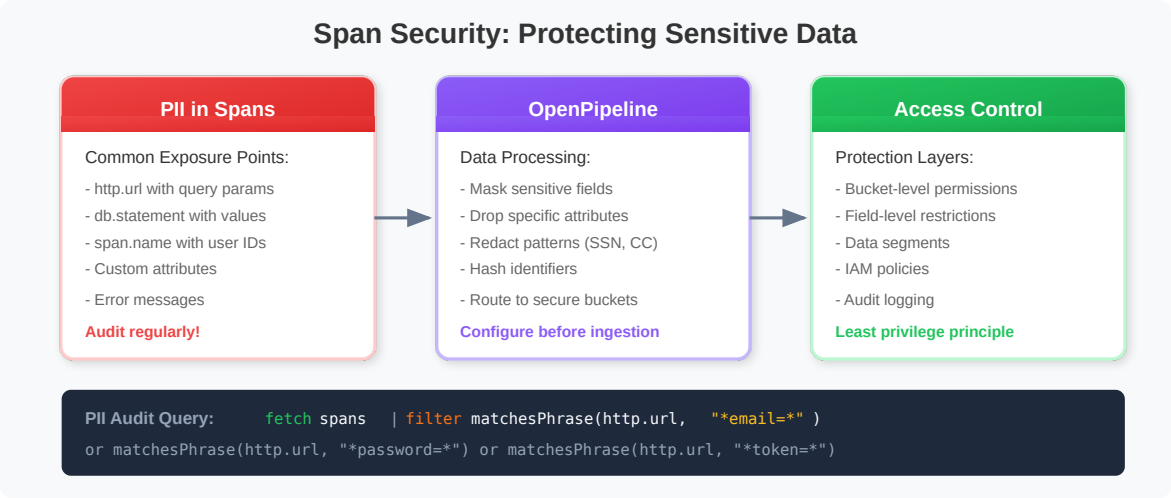
Prerequisites

Before starting this notebook, ensure you have:

- ✅ Completed previous SPANS notebooks (01-05)
- ✅ Understanding of HTTP status codes and security concepts
- ✅ Access to span data containing HTTP attributes (`storage:spans:read`)
- ✅ Familiarity with OpenPipeline basics

1. Understanding Sensitive Data in Spans

Distributed traces can inadvertently capture sensitive information:



Fields to Consider for Protection

Field	Risk	Recommendation
`url.path`	PII in the path	Mask, or group by `http.route`
`url.query` / `url.full`	Tokens and credentials in query parameters	Mask parameter values
`http.request.header.<name>`	Auth tokens/cookies (only headers you configured for capture)	Don't capture them; remove if captured

Field	Risk	Recommendation
`db.query.text` (was `db.statement`)	SQL with user data	Parameterize or mask
`span.events` → `exception.message` / `exception.stack_trace`	Values in exception text	Mask or limit capture
Request attributes / custom attributes	Business data	Evaluate case by case

Exceptions are recorded as **span events**, not span fields: on the validation tenant 6,004 spans in an hour carried an `exception` event with `exception.message` and `exception.stack_trace` inside `span.events`, while a top-level `exception.stacktrace` field has no dictionary row and was set on none.

> **Dictionary:** `url.path`, `url.query`, `url.full`, `db.query.text`, `exception.message`, `http.request.header.__key__` (all `stable`); no row for `exception.stacktrace` or `messaging.payload`, read 10/02/2026 (control: `exception.message` returned a row).

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2. PII Audit Queries

Regularly audit your span data for potential PII exposure. These queries help identify data that may need masking or removal.

Step 1: Find URLs with Email Addresses

```
// Find URLs that might contain email addresses
// Look for @ symbol or URL-encoded %40
fetch spans, from:-1h
| filter isNotNull(url.path)
| filter contains(url.path, "@") or contains(url.path, "%40")
| fields start_time,
         dt.service.name,
         http.route,
         url.path
| dedup http.route
| limit 20
```

Step 2: Find URLs with User Identifiers

```
// Find URLs with potential user identifiers
// contains() is case-sensitive by default, so match case-insensitively
fetch spans, from:-1h
| filter isNotNull(url.path)
| filter contains(url.path, "user", caseSensitive: false) or
        contains(url.path, "email", caseSensitive: false) or
        contains(url.path, "customer", caseSensitive: false) or
        contains(url.path, "account", caseSensitive: false)
| fields dt.service.name,
         http.route,
         url.path
| dedup http.route
| limit 20
```

Step 3: Find Sensitive Query Parameters

```
// Find URLs that suggest credentials or tokens.
// Query strings are carried in url.query (or url.full), NOT url.path — a
path-only check
// never sees "?token=…". Check all three, case-insensitively.
fetch spans, from:-1h
| fieldsAdd url_text = concat(coalesce(url.path, ""), "?", coalesce(url.query,
""), " ", coalesce(url.full, ""))
| filter contains(url_text, "password", caseSensitive: false)
    or contains(url_text, "token", caseSensitive: false)
    or contains(url_text, "secret", caseSensitive: false)
    or contains(url_text, "apikey", caseSensitive: false)
    or contains(url_text, "auth", caseSensitive: false)
| fields dt.service.name, http.route, url.path, url.query
| dedup http.route
| limit 20
```

Step 4: Audit Database Queries for PII

```
// Field names corrected 08/12/2026 — pre-1.0 OpenTelemetry database semconv
names had been
// used throughout, and every one of them is null on Grail spans. They fail
SILENTLY: a filter on a
// non-existent field matches nothing and a summarize groups everything under
null, so these cells
// returned empty or single-null-group results without ever erroring.
//   db.operation         -&gt; db.operation.name    (stable; set on 50,379 of
57,295 db spans)
//   db.statement         -&gt; db.query.text        (stable; 33,463)
//   db.mongodb.collection-&gt; db.collection.name   (stable; 6,912)
//   db.name              -&gt; db.namespace         (stable; 57,281)
// Confirm the catalog for your tenant with:
//   fetch dt.semantic_dictionary.fields | filter startsWith(name, "db.") |
fields name, stability
// Find database queries that might contain sensitive data (contains() is
case-sensitive by default, so match case-insensitively)
fetch spans, from:-1h
| filter isNotNull(db.query.text)
| filter contains(db.query.text, "password", caseSensitive: false) or
        contains(db.query.text, "ssn", caseSensitive: false) or
        contains(db.query.text, "credit", caseSensitive: false) or
        contains(db.query.text, "email", caseSensitive: false) or
        contains(db.query.text, "phone", caseSensitive: false)
| fields dt.service.name,
         db.system,
         db.namespace,
         db.query.text
| limit 20
```

Step 5: Check Error Messages for Data Leakage

```
// Audit error messages for potential data leakage
fetch spans, from:-1h
| filter span.status_code == "error"
| filter isNotNull(span.status_message)
| fields dt.service.name,
         span.name,
         span.status_message
| limit 20
```

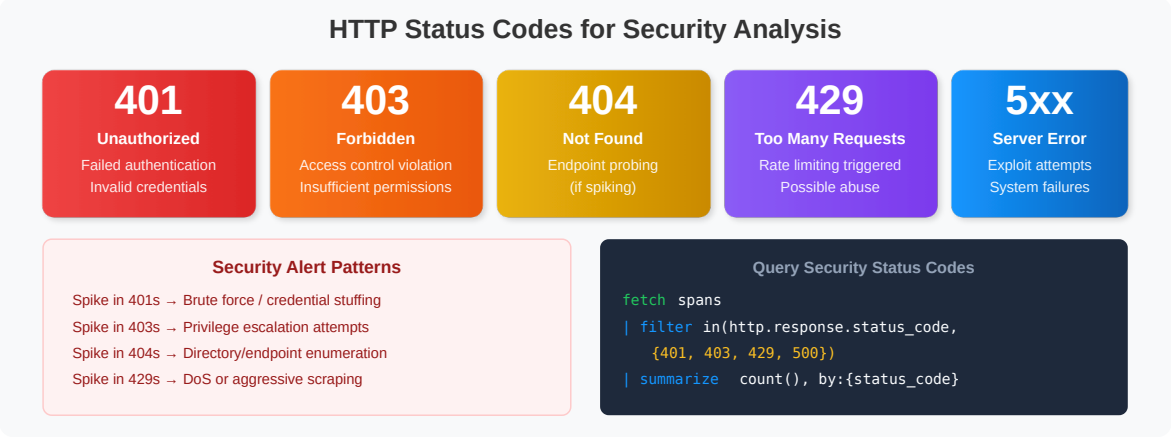
Summary: Which Fields Contain Potentially Sensitive Data?

```
// Identify which potentially sensitive fields are present
fetch spans, from:-1h
| summarize {
    total_spans = count(),
    has_url = countIf(isNotNull(url.path)),
    has_db_statement = countIf(isNotNull(db.query.text)),
    has_exception = countIf(iAny(span.events[][span_event.name] ==
"exception"))
  }
| fieldsAdd url_percent = (has_url * 100.0) / total_spans
| fieldsAdd db_percent = (has_db_statement * 100.0) / total_spans
| fieldsAdd exception_percent = (has_exception * 100.0) / total_spans
```

---

3. HTTP Status Code Security Analysis

HTTP status codes can reveal security-relevant patterns:




```
// Analyze HTTP status code distribution
fetch spans, from:-1h
| filter isNotNull(http.response.status_code)
| summarize {count = count()}, by:{http.response.status_code}
| sort count desc
| limit 20
```

```
// Find security-relevant HTTP errors (401, 403, 429)
fetch spans, from:-1h
| filter in(http.response.status_code, {401, 403, 429})
| summarize {
    count = count()
  }, by:{http.response.status_code, dt.service.name, http.route}
| sort count desc
| limit 50
```

```
// Security status codes over time (detect spikes)
fetch spans, from:-1h
| filter in(http.response.status_code, {401, 403, 429, 500})
| fieldsAdd time_bucket = bin(start_time, 10m)
| fieldsAdd status_category = if(http.response.status_code == 401,
"401_Unauthorized",
                               else: if(http.response.status_code == 403,
"403_Forbidden",
                               else: if(http.response.status_code == 429,
"429_RateLimited",
                               else: "500_ServerError")))
| summarize {count = count()}, by:{time_bucket, status_category}
| sort time_bucket asc, status_category
| limit 200
```

---

4. Authentication Failure Detection

Monitor authentication endpoints for potential brute force or credential stuffing attacks.

```
// Find 401 Unauthorized responses (failed authentication)
fetch spans, from:-1h
| filter http.response.status_code == 401
| fields start_time,
         dt.service.name,
         http.request.method,
         http.route,
         url.path,
         trace.id
| sort start_time desc
| limit 100
```

```
// Count authentication failures by endpoint
fetch spans, from:-1h
| filter http.response.status_code == 401
| summarize {
    failure_count = count(),
    unique_traces = countDistinctExact(trace.id)
  }, by:{dt.service.name, http.route}
| sort failure_count desc
| limit 30
```

```
// Authentication failure rate over time
// contains() is case-sensitive by default, so match case-insensitively
fetch spans, from:-1h
| filter isNotNull(http.response.status_code)
| filter contains(http.route, "auth", caseSensitive: false) or
contains(http.route, "login", caseSensitive: false) or contains(span.name,
"login", caseSensitive: false)
| fieldsAdd time_bucket = bin(start_time, 10m)
| summarize {
    total_attempts = count(),
    failures = countIf(http.response.status_code == 401)
  }, by:{time_bucket}
| fieldsAdd failure_rate_pct = (failures * 100.0) / total_attempts
| sort time_bucket asc
| limit 50
```

```
// High-frequency authentication failures by service
// contains() is case-sensitive by default, so match case-insensitively
fetch spans, from:-1h
| filter contains(span.name, "auth", caseSensitive: false) or
contains(span.name, "login", caseSensitive: false)
| filter span.status_code == "error" or http.response.status_code == 401 or
http.response.status_code == 403
| summarize {
    failure_count = count()
  }, by:{dt.service.name, span.name, http.response.status_code}
| filter failure_count &amp;gt; 10
| sort failure_count desc
```

5. Anomalous Traffic Patterns

Detect unusual patterns that might indicate attacks or misuse.

```
// Find endpoints with unusually high error rates
fetch spans, from:-1h
| filter span.kind == "server"
| filter isNotNull(http.response.status_code)
| summarize {
    total_requests = count(),
    error_4xx = countIf(http.response.status_code &gt;= 400 and
http.response.status_code &lt; 500),
    error_5xx = countIf(http.response.status_code &gt;= 500)
  }, by:{dt.service.name, http.route}
| fieldsAdd error_rate_4xx = (error_4xx * 100.0) / total_requests
| fieldsAdd error_rate_5xx = (error_5xx * 100.0) / total_requests
| filter error_rate_4xx &gt; 20 or error_rate_5xx &gt; 5
| sort error_rate_4xx desc
| limit 30
```

```
// Detect potential enumeration attacks (high 404 rates)
fetch spans, from:-1h
| filter http.response.status_code == 404
| summarize {
    not_found_count = count(),
    unique_paths = countDistinct(url.path)
  }, by:{dt.service.name}
| filter not_found_count &gt; 100
| sort not_found_count desc
| limit 20
```

```
// Find rate limiting events (429 responses)
fetch spans, from:-1h
| filter http.response.status_code == 429
| fields start_time,
         dt.service.name,
         http.route,
         trace.id
| sort start_time desc
| limit 50
```

```
// Unusual access patterns - services with high error rates
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    requests = count(),
    unique_operations = countDistinct(span.name),
    error_rate = (countIf(span.status_code == "error") * 100.0) / count()
  }, by:{dt.service.name}
| filter error_rate &gt; 20
| sort error_rate desc
```

---

6. Sensitive Endpoint Monitoring

Monitor access patterns to sensitive endpoints like admin panels, configuration APIs, and user data endpoints.

```
// Find access to admin or configuration endpoints
// contains() is case-sensitive by default, so match case-insensitively
fetch spans, from:-1h
| filter span.kind == "server"
| filter contains(url.path, "admin", caseSensitive: false)
      or contains(url.path, "config", caseSensitive: false)
      or contains(url.path, "settings", caseSensitive: false)
      or contains(span.name, "admin", caseSensitive: false)
| fields start_time,
         dt.service.name,
         http.request.method,
         url.path,
         http.response.status_code,
         trace.id
| sort start_time desc
| limit 100
```

```
// Monitor data export or bulk operations
// contains() is case-sensitive by default, so match case-insensitively
fetch spans, from:-1h
| filter span.kind == "server"
| filter contains(url.path, "export", caseSensitive: false)
      or contains(url.path, "download", caseSensitive: false)
      or contains(url.path, "bulk", caseSensitive: false)
| summarize {
    access_count = count(),
    unique_traces = countDistinctExact(trace.id)
  }, by:{dt.service.name, http.route}
| sort access_count desc
| limit 20
```

```
// Identify services that might handle regulated data
// contains() is case-sensitive by default, so match case-insensitively
fetch spans, from:-1h
| filter contains(span.name, "health", caseSensitive: false) or
        contains(span.name, "patient", caseSensitive: false) or
        contains(span.name, "payment", caseSensitive: false) or
        contains(span.name, "card", caseSensitive: false) or
        contains(span.name, "billing", caseSensitive: false)
| summarize {
    span_count = count(),
    unique_operations = countDistinct(span.name)
  }, by:{dt.service.name}
| sort span_count desc
```

---

7. OpenPipeline for Data Masking

Use OpenPipeline to mask sensitive data **before** it is stored in Grail.

Masking with a DQL processor

There is no dedicated masking processor. Masking is a **DQL** processor in the **Processing** stage of the spans pipeline, using `replacePattern` with **DPL** patterns (not regex). One processor can mask several fields:

```
fieldsAdd url.path = replacePattern(url.path, "[A-Za-z0-9._%+-]+ '@' [A-Za-z0-9.-]+", "[EMAIL-MASKED]")
| fieldsAdd url.query = replacePattern(url.query, "&lt;&lt;(('token' | 'TOKEN' | 'Token' | 'key' | 'KEY' | 'Key' | 'secret' | 'SECRET' | 'Secret' | 'password' | 'PASSWORD' | 'Password') '=') [^&amp;]+", "[REDACTED]")
| fieldsAdd url.path = replacePattern(url.path, "&lt;&lt;'/' CREDITCARD (&gt;&gt;'/' | &gt;&gt;'?' | EOF)", "[CARD-MASKED]")
```

Each statement was run against sample values on 10/02/2026: `/users/john.doe@example.com/orders` → `/users/[EMAIL-MASKED]/orders`; `token=abc123XYZ&page=2&password=hunter2` → `token=[REDACTED]&page=2&password=[REDACTED]`; `/pay/4111111111111111/confirm` → `/pay/[CARD-MASKED]/confirm`. The card statement is bounded to a whole path segment on purpose: a bare `CREDITCARD` matcher also hits Luhn-valid digit runs **inside** hex identifiers — on the validation tenant it rewrote `/session/c05e365fe5eee852102159234007ec45/element` to `/session/c05e365fe5eee8[CARD]7ec45/element`, and all 30 "card" hits in an hour were session hashes of that kind. `<<(…)` is a lookbehind, so the parameter name stays and only the value is replaced. DPL literals are case-sensitive and there is no case-insensitive flag, so the key list names each casing: with lowercase keys only, `API_KEY=ABCDEF&page=2`, `TOKEN=…` and `Password=…` were left in clear, and with the list above they are masked (verified 10/12/2026). `key` also masks `api_key` and `token` also masks `access_token`, because the lookbehind only checks the characters before `=`. The same reason makes it over-mask `monkey=`, which is the safe direction. `replacePattern` has no back-references, but partial masking ("keep the last four digits") still works with a `>>` lookahead, which checks the trailing digits without consuming them: `replacePattern(url.path, "<<'/' [0-9]{12} >>([0-9]{4} ('/' | '?' | EOF))", "XXXXXXXXXXXX")` turns `/pay/4111111111111234/confirm` into `/pay/XXXXXXXXXXXX1234/confirm` (verified 10/12/2026). Apply the same statements to `url.full` if your spans carry it. For the DPL syntax in depth, see **OPLOGS-08** and **FAQ-15**.

Routing sensitive spans to restricted buckets

Routing chooses a **pipeline**; the pipeline's **Bucket assignment** stage chooses the bucket. Matchers accept `matchesValue` / `matchesPhrase` / comparisons, not `contains()` or `in()`:

Route matcher (first match wins)	Target pipeline	Its Bucket assignment
`matchesValue(dt.service.name, "payment*") or matchesValue(dt.service.name, "auth*")`	`sensitive-spans`	`spans_sensitive`
`matchesValue(deployment.environment, "production")`	`production-spans`	`spans_production`
(default route)	default pipeline	`default_spans`

Spans OpenPipeline configuration is covered in **SPANS-07** and **OPIPE-02**.

Verify Masking Is Working

```
// Verify masking is working — leak check.
// Counts values the masking patterns would STILL change. All three should be
0 after masking.
fetch spans, from:-1h
| summarize {
    total = count(),
    unmasked_email = countIf(replacePattern(url.path, "[A-Za-z0-9._%+-]+ '@'
[A-Za-z0-9.-]+", "") != url.path),
    unmasked_secret_param = countIf(replacePattern(url.query, "&amp;lt;&amp;lt;
(('token' | 'TOKEN' | 'Token' | 'key' | 'KEY' | 'Key' | 'secret' | 'SECRET' |
'Secret' | 'password' | 'PASSWORD' | 'Password') '=') [^&amp;amp;]+", "") !=
url.query),
    unmasked_card = countIf(replacePattern(url.path, "&amp;lt;&amp;lt;'/' CREDITCARD
(&amp;gt;&amp;gt;'/' | &amp;gt;&amp;gt;'?' | EOF)", "") != url.path)
  }
```

---

8. Compliance Considerations

GDPR Compliance

- **Data minimization**: Only collect necessary data
- **Purpose limitation**: Use data only for stated purposes
- **Storage limitation**: Define retention periods

- **Right to erasure**: Plan for data deletion requests

PCI DSS Compliance

- **Do not let** full card numbers reach traces — PCI DSS calls for stored card numbers to be rendered unreadable; confirm the controls that apply to you with your compliance team
- **Mask** cardholder data in all traces
- **Mask** at ingestion rather than relying on storage encryption alone
- **Audit** access to payment-related traces

HIPAA Compliance

- **Protect** PHI (Protected Health Information)
- **Encrypt** health-related span data
- **Restrict** access based on role
- **Audit** all access to healthcare traces

Secure Query Patterns

Use aggregations instead of exposing individual records:

```
// SECURE: Aggregate patterns without exposing individual data
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    request_count = count(),
    unique_routes = countDistinct(http.route),
    error_count = countIf(span.status_code == "error")
  }, by:{dt.service.name}
| sort request_count desc
```

Use `http.route` (pattern) instead of `url.path` (full URL):

```
// SECURE: Use http.route (pattern) instead of url.path (full URL)
// http.route: /users/:id (pattern, no PII)
// url.path: /users/john.doe@email.com (actual value, potential PII)

fetch spans, from:-1h
| filter isNotNull(http.route)
| summarize {
    requests = count(),
    avg_ms = avg(duration) / 1ms
  }, by:{dt.service.name, http.route, http.request.method}
| sort requests desc
| limit 20
```

Limit exposure of trace.id - get just enough to diagnose:

```
// For investigation: Get just enough trace.ids to diagnose
fetch spans, from:-1h
| filter span.status_code == "error"
| summarize {
    error_count = count(),
    sample_trace = takeFirst(trace.id)
  }, by:{dt.service.name, span.name}
| sort error_count desc
| limit 10
```

---

9. Security Audit Queries

Use these queries for regular security audits.

```
// Security summary dashboard: Overall security status
fetch spans, from:-1h
| filter isNotNull(http.response.status_code)
| summarize {
    total_requests = count(),
    auth_failures_401 = countIf(http.response.status_code == 401),
    forbidden_403 = countIf(http.response.status_code == 403),
    rate_limited_429 = countIf(http.response.status_code == 429),
    server_errors_5xx = countIf(http.response.status_code &gt;= 500)
  }
| fieldsAdd auth_failure_rate = (auth_failures_401 * 100.0) / total_requests
| fieldsAdd forbidden_rate = (forbidden_403 * 100.0) / total_requests
```

```
// Security events timeline
fetch spans, from:-24h
| filter in(http.response.status_code, {401, 403, 429})
| makeTimeseries {
    auth_failures = countIf(http.response.status_code == 401),
    forbidden = countIf(http.response.status_code == 403),
    rate_limited = countIf(http.response.status_code == 429)
  }, interval: 10m
```

```
// Services with highest security event rates
fetch spans, from:-1h
| filter isNotNull(http.response.status_code)
| summarize {
    total_requests = count(),
    security_events = countIf(in(http.response.status_code, {401, 403, 429}))
  }, by:{dt.service.name}
| fieldsAdd security_event_rate = (security_events * 100.0) / total_requests
| filter security_events &gt; 0
| sort security_event_rate desc
| limit 20
```

```
// Final audit: Summary of potential security concerns
// contains() is case-sensitive by default, so match case-insensitively
fetch spans, from:-1h
| summarize {
    total_spans = count(),
    spans_with_url = countIf(isNotNull(url.path)),
    spans_with_db_query = countIf(isNotNull(db.query.text)),
    spans_with_errors = countIf(span.status_code == "error"),
    auth_failures = countIf(
        (contains(span.name, "auth", caseSensitive: false) or
contains(span.name, "login", caseSensitive: false)) and
        (span.status_code == "error" or http.response.status_code &gt;= 400))
  }
| fieldsAdd url_percent = (spans_with_url * 100.0) / total_spans
| fieldsAdd db_percent = (spans_with_db_query * 100.0) / total_spans
```

10. Security Checklist

Data Protection

- [ ] Audit spans for PII in URLs, headers, query params
- [ ] Configure OpenPipeline to mask sensitive data
- [ ] Use `http.route` instead of `url.path` when possible
- [ ] Drop or mask database query contents
- [ ] Sanitize error messages and stack traces

Access Control

- [ ] Route sensitive spans to restricted buckets
- [ ] Configure IAM policies for span access
- [ ] Limit who can query raw span data
- [ ] Use aggregations instead of exposing individual records

Compliance

- [ ] Define data retention policies per bucket
- [ ] Document data processing for GDPR Article 30
- [ ] Ensure PCI DSS compliance for payment spans
- [ ] Protect PHI for HIPAA compliance

Monitoring

- [ ] Regular audits for new PII exposure
- [ ] Monitor authentication failure patterns
- [ ] Alert on unusual access patterns
- [ ] Review masking effectiveness periodically

Summary

In this notebook, you learned:

✅ **Understanding sensitive data locations** in span attributes
✅ **PII audit queries** to find emails, tokens, and credentials in spans
✅ **HTTP status code analysis** for security-relevant codes (401, 403, 429, 500)
✅ **Authentication failure detection** to identify potential attacks
✅ **Anomalous traffic pattern detection** for enumeration and abuse
✅ **Sensitive endpoint monitoring** for admin and data access
✅ **OpenPipeline masking** with DQL-processor `replacePattern` statements for emails, card numbers and tokens
✅ **Compliance considerations** for GDPR, PCI DSS, and HIPAA
✅ **Security audit queries** for compliance and reporting
✅ **Security checklist** for ongoing protection

Next Steps

Continue to **SPANS-07: Grail Buckets & OpenPipeline** to learn:
- Understanding Grail bucket architecture
- Configuring OpenPipeline for span processing
- Data routing and retention strategies
- Access control for span data

References

- Trace semantic conventions (DT docs)
- Processing stage (DT docs)
- DQL matcher in OpenPipeline (DT docs)
- Dynatrace Pattern Language (DT docs)

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